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Talal Aladaileh
School of Systems Science and Industrial Engineering, Binghamton University, Binghamton, NY, USA, taladaileh@binghamton.edu

Donald Stephens
School of Systems Science and Industrial Engineering, Binghamton University, Binghamton, NY, USA, dstephens1@binghamton.edu

Mallak Alqaisi
School of Systems Science and Industrial Engineering, Binghamton University, Binghamton, NY, USA, malqaisi@binghamton.edu

Congyu Wu
School of Systems Science and Industrial Engineering, Binghamton University, Binghamton, NY, USA, congyu.wu@binghamton.edu

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Solving Wordle Using Information Theory

Talal Aladaileh [1*†], Donald Stephens [1†], Mallak Alqaisi [1†], and Congyu Wu [1]

[1] School of Systems Science and Industrial Engineering, Binghamton University, Binghamton, NY, USA

* taladaileh@binghamton.edu

† These authors contributed equally to this work

Abstract

Wordle, a popular word-guessing game, challenges players to identify a five-letter secret word through iterative guesses and feedback on letter placement. The players must figure out the secret word within six guesses. After each guess, the letters will be color-coded based on different criteria. Optimizing the choice of guesses is critical for maximizing success within the limited attempts allowed. In this study, the application of Shannon entropy is explored as a strategy for selecting words that maximize information gain at each step of the game. By quantifying the uncertainty reduction achieved by potential guesses, this method prioritizes words that are likely to narrow the solution space most effectively. The experimental results demonstrate that entropy-based word selection improves performance compared to a heuristic approach based on selecting words by letter distribution, providing a systematic framework for decision-making in Wordle. This study can be used as a baseline for implementing Shannon entropy in different games.

1. Introduction

Wordle is a popular online single player game developed by Josh Wordle, where players attempt to identify a secret word by making five-letter word guesses. The game is won if the player can guess the secret word in six guesses or less; otherwise, the game is lost. Following each guess, players receive feedback indicating which letters are absent from the word (highlighted as gray), which letters are present but incorrectly positioned (highlighted as yellow), and which letters are both correct and in the proper position (highlighted as green). Using this information, players refine their guesses by eliminating incorrect options and selecting new possibilities for subsequent attempts.

Beyond being well-known as a simple word-guessing game, Wordle can be looked at as a dynamic feedback system in which each guess will provide information that will affect the subsequent guesses. Through this continuous feedback, the state of the game develops as players learn from clues and eliminate possibilities, which reduces the uncertainty with each round. This uncertainty is quantified using entropy; as feedback narrows down the possible solutions, the entropy (uncertainty) of the game state decreases, moving the solution from disordered state initially to more organized state. In this way, information theory offers a clear framework for analyzing the decision-making process works and adapts with each guess in Wordle.

The game opens with an empty 6×5 grid as shown in Figure 1, where players choose an initial word as their first guess. After submitting a guess, the game highlights letters as previously described to give the player clues. For example, if the secret word is "plant" and the word "slate" is played, the game would highlight the letters of slate as: gray; green; green; yellow; gray. Here, letter S and E receive a gray highlight because they are not in the secret word. Letters L and A each receive green highlights as they are in the secret word and correctly positioned. The letter T receives a yellow highlight because it is in the secret word but in the wrong position. Careful inspection of the highlighted letters enables the player to eliminate potentially incorrect words ultimately lowering the search space.

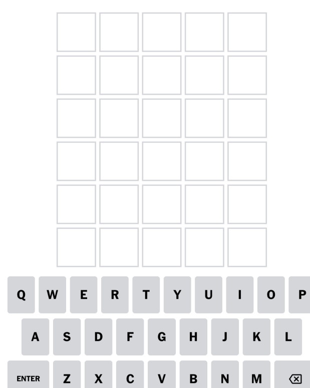

Figure 1. Opening screen of Wordle

In this paper, we explore how Shannon entropy can be used to reduce the search space significantly on each step, resulting in a more efficient strategy to win Wordle on average.

In 1948, Claude Shannon published a paper called "A Mathematical Theory of Communication" [1]. This presented the opportunity for information to be treated as a well-defined and measurable quantity. Information defines definite, unbreach-

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able limits on precisely how much information can be communicated between any two components of any system.

Information is usually measured in bits, and one bit of information allows someone to choose between two equiprobable alternatives. The word bit is derived from binary digit, but a bit and a binary digit are fundamentally different types of quantities. A binary digit is the value of a binary variable, whereas a bit is an amount of information. Entropy is the expected average of information. The entropy gives a measure of the uncertainty of the random variable. The larger the entropy, the less a priority information one has on the value of the random variable.

If X is a random variable, then Information and Entropy, shown in Equations 1 and 2, can be defined, respectively, as:

$$I\left(x\right) = -\log_2\left(P\left[x\right]\right), \quad x \in X \tag{1}$$

$$H\left(X\right) = E\left[I\left(X\right)\right] = -\sum_{x \in X} \log_2\left(P\left[x\right]\right) P\left[x\right] \tag{2}$$

where $P\left[\ldots\right]$ denotes the probability, and $E\left[\ldots\right]$ denotes expectation. Additionally, by convention we let $0\log_2\left(0\right) = 0$.

The specific research questions this paper aims to address is: can information theory be used to identify the best starting word and subsequent words in Wordle to reduce the total number of guesses needed on average to win the game? This question arises from the recognition that entropy provides a quantitative measure of uncertainty reduction, making it a valuable framework for decision making in feedback-driven contexts like Wordle. Building on this idea, we developed an entropy-based strategy to solve Wordle, aiming to demonstrate its effectiveness in reducing the solution space relative to a letter distribution strategy.

2. Related Literature

The study of Wordle optimization has shown some different methodologies, ranging from computational algorithms to human-centered strategies. Prebreza et al. [2] showed that entropy-based solvers outperform simple frequency-based heuristics in achieving high accuracy and efficiency consistently. The approach used was only limited to algorithmic comparisons without having large-scale validation. Additionally, using features such as frequency, word entropy, and repetition counts, Xin et al. [3] studied Wordle outcomes from a predictive point of view to classify how the solution is difficult using XGBoost and decision trees.

Mishra [4] investigates in her research the role of entropy as a metric to improve the guesses in Wordle. The study shows how entropy can determine the information gained from each guess by considering the chance the patterns can happen and their possibility to decrease the solution space. The study states that entropy exceeds easier metrics like average size in effectiveness. By putting together Wordle as an information-theoretic problem, Mishra evolves an algorithm that achieves higher accuracy by lowering the guess counts, presenting a computationally realistic and strong approach to the game. The study also shows entropy's usefulness in giving a systematic and quantitative structure for making decisions in this game, demonstarting the wider applications of information theory in puzzle-solving frameworks. The limitations in this study are that the algorithm used can't distinguish between closely related words, which increases the guess count for each case, it also only evaluates one guess forward, which reduces the potential for more strategic combinations.

Gui et al. [5] expand Wordle optimization research by combining mutual information to rank words into "easy", "normal", and "hard" categories. Using K-means clustering, the authors collect words based on their difficulty, which are confirmed through Monte Carlo simulations and dynamic programming. This research initiates a novel classification method that boosts strategy formulation by matching approaches to word difficulty. The combination of mutual information provides a new aspect to understanding Wordle, moving beyond only optimization to label variation in word difficulty. This study closes the gap between theoretical models and practical applications, encouraging bigger engagement and customized plans for players, but it also has some limitations, as it only focuses on 2,309 pre-identified words, which may not generalize to broader word pools. It also assumes players use uniform guessing probabilities without leveraging strategic insights.

Gomez and Puga's research [6] investigates entropy-driven ways for solving Wordle, comparing greedy and genetic algorithms. The study assesses the computational effectiveness and accuracy of those different algorithms in driving into the solution space. While greedy algorithms focus on instant developments, genetic algorithms imitate evolving processes to iteratively enhance guesses. The authors focus on entropy's part in assessing uncertainty reduction, displaying its advantages as a decision-making tool. However, they also consider the trade-offs between computational cost and solution accuracy, especially with genetic algorithms needing more repetitions for convergence. The study highlights how resourceful entropy can be in conducting algorithmic choices, helping to the growing body of research on Wordle optimization. The limitations for this study fall in that it assumes uniform word probability, ignoring real-world biases in word selection, and in some heuristic challenges where greedy algorithms struggle to balance entropy maximization with real solution frequencies.

De Silva [7] utilizes character frequency analysis to find the best starting words for Wordle. The study builds a classified list of words based on their statistical probability of having high-frequency letters, suggesting combinations like “raise”, “clout”, and “nymph” for the most letter coverage. The research issues a heuristic method that simplifies the initial word option by focusing on the frequency distribution of letters across Wordle’s word list. The importance of character frequency provides an uncomplicated yet effective strategy for players, balancing clarity with possibility. However, the study’s dependency on frequency-based heuristics might miss the significant information benefit achieved through more complex metrics like entropy, and it relies on a curated word list that may differ from Wordle’s actual word pool, which might lead to discrepancies in recommendations.

The previous research papers jointly explore various procedures to improve Wordle strategies, from entropy-based methods to statistical analysis, yet most studies failed to validate their findings experimentally while ignoring real-world player adaptation methods. Our research fills this gap by testing an entropy-based gameplay method against letter distribution strategies to prove that entropy reduces the number of guesses needed to solve the game.

3. Data

Wordle is offered as a daily puzzle with one unique secret word every day of a given year. There are 12,972 valid five-letter words present in Wordle’s dictionary, of which 2,315 words are candidates for the daily puzzles. A copy of the 12,972 words was obtained which is made freely avaiable online by Wordle’s authors.

Figure 2 shows the plot of the distribution of words by unique letters. For example, the number of unique words that contain the letter ”a” is 912. The plot illustrates that the majority of the five-letter words contain at least one “e”. Words with duplicate letters, such as “steel” are only counted once for each letter (in this case the letter “e”).

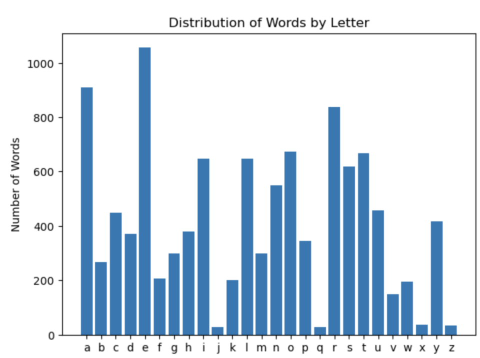

Figure 2. Distribution of words by letter

This plot will be helpful later when a gameplay strategy is discussed based on letter distribution to select a word as a guess. Overall, the majority of the words are composed of letters “a”, “e”, and “r”. Conversely, less words contained letters “j”, “q”, “x”, and “z”. Adding the number of words that letters from ”a” to ”z” appeared in will not equal 12,972 as words can fall into more than one letter bucket (e.g., “abode” falls into 5 letter buckets).

The relative percentage of words by number of unique letters was also considered, observing that just under 70% of words have five unique letters as shown in Figure 3. As the number of unique letters decrease there is a significant decrease in the relative percentage of words.

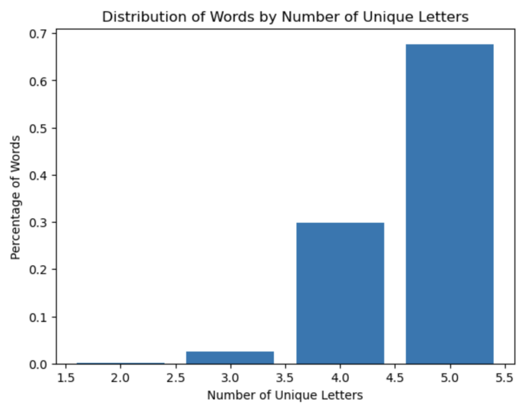

Figure 3. Distribution of words by number of unique letters

Though Wordle is offered as a daily puzzle online, collecting any gameplay data or historical data on word guesses overtime by Wordle users were not doable. This did not cause any issues as historical data was not required for the analysis or assessment.

4. Methods

In this section, the entropy based gameplay strategy and a baseline gameplay strategy based on selecting words by letter distribution will be discussed. The goal was to develop a gameplay strategy based on Shannon entropy to play Wordle. Later in this study, a comparison between the entropy based strategy against the selected baseline gameplay strategy will be shown.

Recall that once a Wordle game starts, color highlighting is used by Wordle to denote whether letters of the guessed word belong to the secret word. In particular, the three different states a letter can take each time a word is played: “gray”, ”yellow”, and “green” were described. Since there are five positions, each taking on three different states, there are ultimately 243 letter state possibilities for each word played as described in Figure 4 and Figure 5.

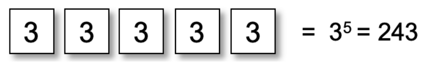

Figure 4. Number of letter state possibilities

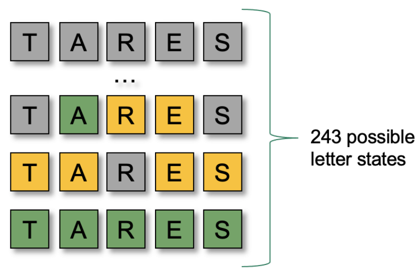

Figure 5. Excerpt of letter states for word "tares"

The probability of each letter state possibility must be calculated first. To do this, the number of words in the 12,972 five-letter word list that match the pattern of the given letter state possibility was counted and divided by 12,972.

Now let "Words" be the set of 12,972 five-letter words, and $\sim$ to denote that word w matches the pattern s.

$$P\left[s\right] = \frac{|\{w \sim s \ \mid \ w \in Words\}|}{|\{w \ \mid \ w \in Words\}|} \tag{3}$$

Equation (3) provides a probability for each letter state, for a given word. As such, there will be 243 probabilities that sum to 1. If the probabilities are sorted in increasing order, the probability mass distribution can be illustrated as depicted in Figure 6. The x-axis in Figure 6 refers to a letter state and the y-axis is the probability of that letter state occurring. Sorting the probabilities in increasing order is an arbitrary decision and doesn't change the interpretation. Notice that some states would have a very low probability where others would be relatively higher. A prime example of a letter state with an extremely low probability would be a letter state of all "green" highlighted letters, in this particular example there should only be 1 word that matches such a pattern, hence the probability would be $1/12972$. A prime example of a letter state with a relatively high probability would be a letter state of all "gray" highlighted letters.

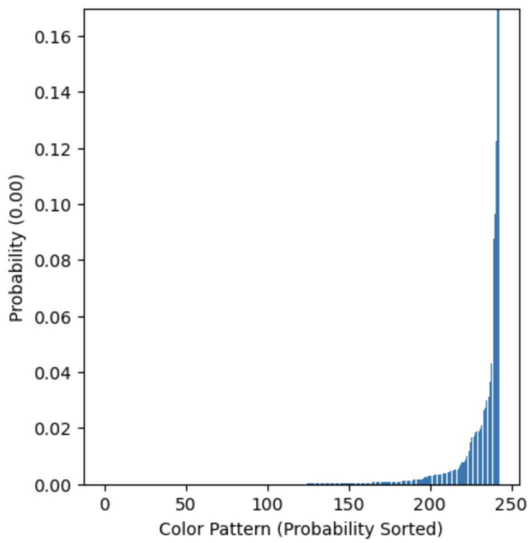

Figure 6. Example of probability mass distribution of letter states

Next, we calculate the information content for each possible letter-state outcome of a guess. Here, "information" (in bits) measures how much uncertainty is reduced by observing a particular outcome. The analysis of all gray letters in the outcome reveals important data because it removes numerous potential solutions from consideration. Using the definition of information (Equation 1) for each outcome s:

$$I(s) = -\log_2 (P[s]) \tag{4}$$

The entropy $H(S)$ of the guess is then the expected information over all possible outcomes. This result follows from:

$$H(S) = E[I(S)] = -\sum_{s \in S} \log_2 (P[s]) \, P[s], \tag{5}$$

where S includes every single possible letter position result when guessing a word, there are 243 such outcomes in Wordle. Playing that particular word reveals an amount of information we can expect; that expectation is measured in bits.

The literal interpretation of entropy is the expected amount of bits gained by playing the corresponding word. The term "expected" refers to statistical expectation, it does not imply that this exact amount of information will be obtained. The actual amount of information gained depends on the state of the letters highlighted by Wordle after a word is played; at this point it would be known which of the 243 possible letter states the guessed word resulted in. The reason this cannot be known ahead of time is because the secret word is truly a secret and the game does not give any prior clues to the secret word, it can only be found by playing words.

To provide some more context on the probability distribution of the letter states for a given word, two random words were selected, “audio” and “tares”, and overlaid their probability distributions, illustrated in Figure 7. Notice that the probability distribution for “audio” is taller than that of “tares”. Since information has an inverse relationship with probability, what the distribution is denoting is that higher information has lower probability and vice versa. Hence, by choosing words with higher expected information (i.e., entropy), rarer situations are encountered and substantially more information is gained.

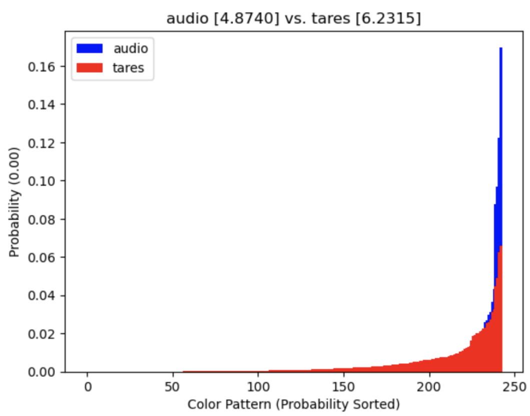

Figure 7. Probability distributions of two words

Once the entropy for each of the 12,972 words is known, the words were sorted in descending order such that the topmost word has the highest entropy. This topmost word has the most expected information that can be gained by playing that word. An excerpt of the topmost 10 words based on the calculations is shown in Figure 8.

word	expected entropy
tares	6.231549
lares	6.211607
rales	6.188127
rates	6.153093
soare	6.138808
nares	6.138196
reais	6.119933
teras	6.117745
arles	6.113117
tales	6.111838

Figure 8. Top 10 words with highest expected information (entropy)

The gameplay strategy can be summarized as follows::

- Pick and play the word with the highest entropy as the first guess (e.g., play “tares”). Wordle will then communicate the letter states of playing that word (e.g., “green”, “gray”, “gray”, “yellow”, “gray”).
- Using the letter state communicated by Wordle, get all of the words for that letter state for the played word. Recall that the probabilities of 243 letter states for each of the 12,972 words were calculated. Also, the words that matched those letter states were saved. Hence, the list of words for ”tares” that matched the state: ”green”, ”gray”, ”gray”, ”yellow”, ”gray” is obtained in this example.
- Pick and play the word in the new list with the highest entropy as the next guess.
- Continue this pattern until the game ends (i.e., find the secret word or you exhaust all of your six guesses).

Figures 9 and 10 serves as an illustrative example. The entropy based gameplay strategy described above was tested. It’s noticeable that using the entropy based gameplay strategy the game was won in four guesses. The process started with “tares” as the initial guess, followed by playing the word with the highest entropy at each step for subsequent guesses.

Figure 9. Wordle screen

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```
Word Guessed: tares [6.2315]
What was the resulting pattern:  0
- Game state from word: ['Gray', 'Gray', 'Gray', 'Gray', 'Gray']
- Actual Bits: 8.6511
- Next word options: 402 (Showing top 10)
```

	word	expected entropy
0	noily	4.931624
1	colin	4.918668
2	doily	4.900264
3	poind	4.830866
4	lound	4.819427
5	poilu	4.789257
6	nould	4.782810
7	nicol	4.766419
8	dolci	4.749271
9	login	4.745040

```
What word to play?  noily

Word Guessed: noily [4.9316]
What was the resulting pattern:  3
- Game state from word: ['Gray', 'Gray', 'Gray', 'Yellow', 'Gray']
- Actual Bits: 2.8074
- Next word options: 7 (Showing top 10)
```

	word	expected entropy
0	mulch	4.067046
1	clump	4.038915
2	gulph	4.000907
3	gulch	3.982489
4	pluck	3.965721
5	plumb	3.931618
6	flump	3.766270

```
What word to play?  mulch

Word Guessed: mulch [4.0670]
What was the resulting pattern:  117
- Game state from word: ['Yellow', 'Yellow', 'Yellow', 'Gray', 'Gray']
- Actual Bits: 1.0000
- Next word options: 2 (Showing top 10)
```

	word	expected entropy
0	plumb	3.931618
1	flump	3.766270

```
Congratulations!
```

Figure 10. Entropy based gameplay strategy

For the baseline, a strategy was selected comparable to the natural way many people play Wordle, that is, playing words with letters that are found in most five-letter words as shown in Figure 2. This strategy works as follows:

- Pick and play any word that contains the letters “a”, “e” and “r”, since those letters cover most words in the 12,972 list of words. Coincidentally, “tares” is an example of such a word. Wordle will then communicate the letter states of playing that word (e.g., “green”, “gray”, “gray”, “yellow”, “gray”).

- Collect the letters that get highlighted “green” and “yellow” as required letters, and letters highlighted as “gray” as invalid letters. If the number of required letters is less than five, select any word that contains the current set of required letters, does not contain any of the invalid letters, and contains a letter, not in the required letters list nor the invalid letters list, that has the next highest word frequency.

- Continue this pattern until the game ends (i.e., until the secret word is found or all six guesses are exhausted).

An excerpt of this strategy at work was included, assuming the secret word was "crypt" and the starting guess was "tares", as shown in Figure 11. Notice that the game was won in four guesses.

```
Guessed word: tares | Iteration: 0
-> letter_modes: [('o', 242.0), ('i', 214.0), ('u', 177.0), ('r', 169.0), ('y', 153.0), ('l', 151.0), ('
t', 148.0), ('n', 147.0), ('c', 125.0), ('h', 121.0), ('d', 100.0), ('g', 91.0), ('m', 79.0), ('p', 77.
0), ('b', 72.0), ('f', 60.0), ('w', 49.0), ('k', 48.0), ('v', 15.0), ('j', 9.0), ('z', 8.0), ('x', 6.0),
('q', 5.0)]
--> Found 25 words
 --->
{'throb', 'froth', 'court', 'motor', 'intro', 'trout', 'throw', 'tutor', 'forth', 'robot', 'troop', 'wort
h', 'broth', 'orbit', 'north', 'troll', 'torch', 'forty', 'turbo', 'front', 'droit', 'thorn', 'grout', 't
umor', 'rotor'}
----------------------------------------

Guessed word: throb | Iteration: 1
-> letter_modes: [('i', 101.0), ('u', 73.0), ('y', 73.0), ('l', 62.0), ('n', 47.0), ('r', 45.0), ('t', 3
8.0), ('c', 36.0), ('d', 34.0), ('g', 32.0), ('p', 31.0), ('f', 28.0), ('m', 27.0), ('k', 26.0), ('v', 9.
0), ('w', 9.0), ('z', 5.0), ('q', 4.0), ('j', 3.0), ('x', 1.0)]
--> Found 8 words
 --->
{'fruit', 'drift', 'twirl', 'flirt', 'print', 'dirty', 'trick', 'fritz'}
----------------------------------------

Guessed word: fruit | Iteration: 2
-> letter_modes: [('y', 4.0), ('r', 3.0), ('l', 2.0), ('p', 2.0), ('c', 1.0), ('d', 1.0), ('g', 1.0), ('
m', 1.0), ('t', 1.0), ('w', 1.0)]
--> Found 1 words
 --->
{'crypt'}
----------------------------------------

Guessed word: crypt | Iteration: 3
('crypt', 3)
```

Figure 11. Example of baseline strategy for secret word "CRYPT"

5. Results

The entropy-based plan was evaluated next to the basic letter-distribution plan. This was checked in two ways:

- A few secret words were picked, then both strategies were tried on each secret word. The chart puts each case side by side. In some words the baseline found the solution in fewer guesses, in others both strategies performed equally, and in many cases the entropy-based strategy won with fewer guesses. Figure 12 shows some test samples. The samples cover a baseline win, a tie, and an entropy win.

Secret word: ABODE

Guesses	Baseline	Entropy
1	TARES	TARES
2	PLANE	LIANE
3	AWOKE	COMAE
4	ABODE	APODE
5		ABODE

Winner: Baseline

Secret word: TARES

Guesses	Baseline	Entropy
1	TARES	TARES

Winner: TIE

Secret word: SLATE

Guesses	Baseline	Entropy
1	TARES	TARES
2	LEAST	SLATE
3	LEAST	
4	LEAST	
5	LEAST	
6	LEAST	

Winner: Entroy

Figure 12. Contrasting both strategies on random word sample

- Full simulation: An exhaustive simulation was conducted using all 2,315 possible Wordle solution words as secret words. The game was played with every secret word, and both strategies were used to guess the word. The same first guess, ”tares”, the highest-entropy word, was used for both strategies each time, and play continued until the word was found or six guesses were used. The number of guesses made by each strategy was then counted, and each game was marked as a win or a loss. The results were then examined to assess performance across games. This can be seen in Figures 13 and 14. The entropy-based plan performed better than the baseline in most cases. In the full test, the entropy strategy guessed almost every secret word within six tries, achieving a success rate of over 99%. The frequency-based method guessed the word about 90% of the time. On average, the entropy method found the secret word in fewer tries than the baseline. These results show that the entropy-driven approach performs better than the letter-frequency approach.

6. Conclusion

Significant evidence was found that the entropy-based gameplay strategy:

- can identify the best starting word and subsequent words in Wordle to reduce the total number of guesses needed on average to win the game.
- on average can outperform a strategy based on making guesses by letter distribution.

6.1. Key Observations

The letter based gameplay strategy can get caught up in an unending cycle if multiple words contains the same five letters. As an example, consider the letters

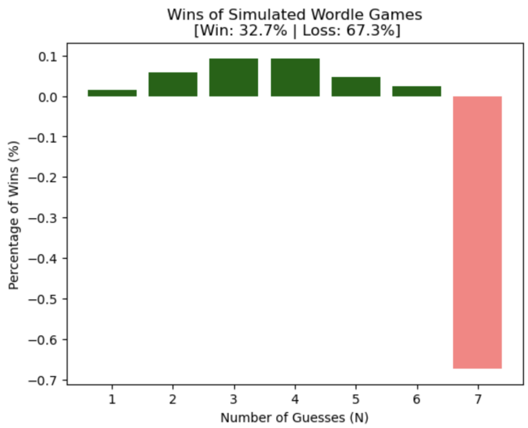

Figure 13. Simulated baseline strategy

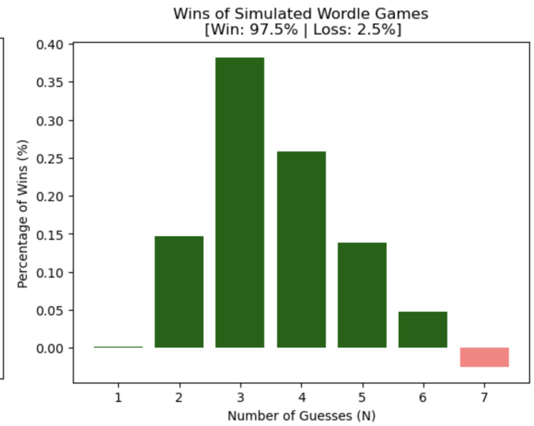

Figure 14. Simulated entropy strategy

of “stale”. This strategy could get stuck on one of the following words: “least”, “stale”, “slate” or “steal”.

Though the entropy-based gameplay strategy has a high percentage of wins on average, it does not win every game 100% of the time.

6.2. Known Limitations

The research needs to evaluate the entropy-based strategy against both the baseline heuristic and other sophisticated Wordle solvers developed during the last few years. Notably, exact search algorithms have achieved superior performance; for example, an optimal dynamic programming approach can guarantee solving any Wordle in at most five guesses, with an average of about 3.42 guesses using the best starting word [8]. This mathematically optimal strategy outperforms the greedy entropy method in terms of both consistency and efficiency. Likewise, other researchers have explored machine-learning and search-based solutions. The solution space of Wordle becomes accessible through three different methods, including neural-network-guided reinforcement learning for policy improvement, genetic algorithms, and advanced search methods [3, 6]. These complex algorithms function to reduce the number of required guesses and ensure victory within six attempts. The entropy-based solver functions as a one-step greedy system because it lacks multi-round prediction features or experience-based learning from past matches, which simplifies calculations but leads to occasional difficult situations that require sophisticated planning capabilities. The research results align with previous studies demonstrating that Wordle players achieve better results using entropy-based methods rather than simple frequency-based rules [2, 4]. The entropy-based method operates as an information-theoretic approach that creates a quantitative prediction

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system producing better results than simple heuristics while remaining more interpretable than state-of-the-art peak-performance solvers.

The entropy-based strategy requires a list of five-letter valid words before the strategy can be employed. It also does not employ any additional optimization nor heuristics; it just takes the word with the highest expected information (i.e., entropy) during each guess. Moreover, the current strategy always takes the total 12,972 words as the basis for any calculation, thus it does not take into account the words that already served as answer key previously, as the game gets played day by day.

The research demonstrates Wordle functions as a basic representation of complex adaptive systems because players use information exchange and feedback to achieve a more organized state of the game. The entropy-based strategy both enhances gameplay performance and shows how systems with information-based adaptation follow the same patterns as complex systems that develop through feedback processes and uncertainty reduction. The discovered link between information theory concepts and system adaptation demonstrates that these principles effectively describe how complex systems learn and adapt through self-organization for maximum efficiency.

6.3. Future Work

In the future, within the entropy-based gameplay strategy where the probability of each letter state is calculated, words could be weighted such that those used more frequently in conversational English receive higher weights than other words. The current strategy assumes that all words fitting a particular letter state pattern are equally weighted.

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